Inverse Kinematics Approximation for 6-DOF Anthropomorphic Robots Using Machine Learning Techniques

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***Abstract*—Traditional inverse kinematics (IK) for 6-Degree-of-Freedom (6-DOF) anthropomorphic robots relies on complex, computationally expensive analytical or iterative Jacobian-based solvers. These mathematical approaches often struggle with kinematic singularities and multi-modal configurations. This project implements a machine learning-based IK approximation system that represents spatial end-effector coordinates as numerical feature vectors and retrieves the corresponding joint angles using traditional algorithms, bypassing heavy trigonometric computation. The system is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared metrics across multiple regression models (Multiple Linear Regression, Support Vector Regression, Decision Trees) to determine the configuration that yields the most accurate joint mappings. The results are expected to demonstrate that non-linear, kernel-based regressors can meaningfully reduce IK computation time while maintaining acceptable precision thresholds for non-critical robotic tasks.**

***Index Terms*—Inverse Kinematics, Support Vector Regression, Machine Learning, Robotics, 6-DOF, Decision Trees**

I. Introduction and Motivation

Inverse kinematics problems (IKP) are ubiquitous in robotics for improved robot control, but the high non-linearity, complexity, and equation coupling of a general six-axis robotic manipulator pose substantial challenges. In fact, up to 16 inverse kinematics solutions can be expected for a general six degrees of freedom (6-DOF) robot. While analytical methods are efficient in solving the IK equation, they frequently suffer from low accuracy, poor real-time performance, and difficulty ensuring stable operations in dynamic environments [1].

The primary motivation for this project stems from the need for a lightweight, data-driven approximation approach that does not require the iterative matrix inversions demanded by traditional Jacobian pseudo-inverse methods, which can suffer from slow convergence rates and stability problems near singularities. Support Vector Regression (SVR) and Decision Trees are well suited to this setting: given a vector representation of an end-effector target, they map the coordinates directly to joint space, requiring no complex analytical derivations and executing inference in constant time.

II. Problem Statement

Robotic controllers frequently need to resolve target end-effector poses into valid actuator commands. The problem addressed in this project is twofold: (1) representing 6D target poses (x, y, z, roll, pitch, yaw) in a feature space that can be accurately mapped to a continuous 6-dimensional joint angle space, and (2) minimizing the predictive error of multi-output regression models over that feature space, while quantifying how the choice of non-linear transformations affects approximation quality.

III. Objectives

The specific objectives of this project are:

1) To collect and pre-process a dataset of 6-DOF robot arm kinematics suitable for supervised learning.
2) To represent each target spatial configuration as a normalized numerical feature vector.
3) To implement a multi-output regression engine utilizing Multiple Linear Regression, SVR, and Decision Trees.
4) To evaluate approximation quality using MAE, RMSE, and R^2 on a held-out test set.
5) To compare linear combinations against non-linear Radial Basis Function (RBF) kernels to identify the configuration that yields the most accurate joint predictions.

IV. Related Work

Machine learning and metaheuristic algorithms have increasingly been utilized as a competitive baseline for kinematic approximation tasks, framing the inverse kinematics problem as an optimization challenge to reduce reliance on cumbersome numerical techniques [2]. Recent works have successfully explored neural networks [5], genetic algorithms [7], and hybrid approaches [4] for simulation strategies in robotic manipulators. Traditional regression approaches, such as Support Vector Machines [9] and regression trees, remain highly competitive in settings with limited training data or strict real-time inference constraints [12].

V. Proposed Methodology

The overall experimental workflow is: Raw Kinematic Data $\rightarrow$ Data Normalization $\rightarrow$ Model Training $\rightarrow$ Multi-Output Prediction $\rightarrow$ Evaluation

Each spatial target X is represented as a vector consisting of position and orientation:

$$X_i = [x, y, z, \theta_{roll}, \theta_{pitch}, \theta_{yaw}]$$

For the baseline Multiple Linear Regression, the relationship to the joint angle vector Y is modeled as:

$$Y = X\beta + \epsilon$$

To capture the non-linear trigonometric relationships inherent in robotic joints, Support Vector Regression (SVR) maps the input vectors into a higher-dimensional feature space using an RBF kernel. The kernel similarity between two spatial vectors is computed as:

$$K(X_i, X_j) = \exp(-\gamma||X_i - X_j||^2)$$

To handle the 6 individual joint angles (q_1 through q_6), the regressors are wrapped in a multi-output meta-estimator strategy, training independent models or decision branches for each degree of freedom.

VI. Evaluation Framework

Evaluation is divided into two categories.

A. Approximation Accuracy

For a held-out test set of spatial coordinates, the predicted joint angles are compared against the ground truth labels. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) are computed for each individual joint j:

$$RMSE_j = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_{ij} - y_{ij})^2}$$

B. Algorithm Comparison

The relative prediction quality of Multiple Linear Regression, SVR (with RBF kernel), and Decision Tree Regressors is compared under identical feature representations to isolate the effect of the algorithmic mapping from the effect of feature engineering.

VII. Implementation Plan

A. Phase 1: Data Collection and Pre-processing

- Acquire the ABB IRB2400 Inverse Kinematics dataset.
- Normalize spatial coordinates and joint angles to standardize variance across the dataset.

B. Phase 2: Feature Engineering

- Construct the input feature matrices and output target matrices.
- Utilize K-Means clustering to identify distinct configuration modes if multi-modality is present.

C. Phase 3: Model Implementation

- Implement the multi-output regression pipelines using Python and scikit-learn [12].
- Support multiple algorithmic architectures for direct comparison.

D. Phase 4: Evaluation and Analysis

- Compute MAE, RMSE, and R^2 across all joint angles.
- Analyze how hyperparameter tuning affects approximation accuracy [14].

VIII. Expected Outcome

This project is expected to produce a working multi-output machine learning pipeline that accepts a 6D spatial target and returns the required 6 joint angles for an anthropomorphic robot. The results are expected to show that non-linear models like SVR (RBF) and Decision Trees vastly outperform baseline linear regression on kinematic data, and that traditional ML approaches provide a reasonable accuracy-to-inference-speed trade-off relative to computationally heavy analytical solvers.

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